INCH-POUND

MIL-P-45819D
12 June 1992
SUPERCEDING
MIL-P-45819C
15 August 1985

MILITARY SPECIFICATION

PANS, DRAIN, AUTOMOTIVE

This specification is approved for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers two sizes of drain pans used for collecting oil being drained from automotive vehicles.

1.2 Classification.

1.2.1 Sizes. The drain pan shall be of the following minimum sizes, as specified (see 6.2):

Size I -- four gallon capacity
Size II-- six gallon capacity

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910
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PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publishing and Printing Service Office Bldg. 4D, NPM-DODSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shallfurnish one or more drain pans for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The drain pan shall be new and of the manufacturer's latest design meeting the requirements specified herein. The pan shall be shallow receptacle equipped with an anti-splash baffle for pouring. Each pan shall include handles for lifting, pouring and positioning the pan under an automotive vehicle to receive oil drainage. The design of non-metallic pans shall include ribbed bottoms and sides to provide strength and rigidity suitable for transporting when filled to capacity.

3.2.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt, from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3 Construction. All parts and materials used to manufacture the pan shall be free of defects. Patching processes shall not be used to reclaim parts or surfaces. Surfaces shall be clean and free of extraneous materials. Edges shall be rounded.

3.4 Welding, brazing, and soldering. Welding, brazing, and soldering, when applicable, shall be of a quality which shall sustain all requirements brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.5 Performance and product characteristics. Drain pans shall be capable of receiving and conveying drained engine oil without splashing or leakage when filled to the capacities specified in 3.5.1. The drain pan shall incorporate provisions which shall allow one operator, in a standing position, to push or pull the pan on a shop floor. Pans shall also include lifting and pouring capabilities.

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3.5.1 Capacities. The size I drain pan shall have a capacity of no less than four gallons and the size II pan shall have a capacity of no less than six gallons.

3.5.2 Leakage. Drain pans shall retain a mixture of water and detergent for not less than 15 minutes, without visible evidence of leakage, after being pulled a distance of no less than 15 feet across a concrete floor. The detergent and water mixture shall consist of not less than one measured cup of liquid detergent, containing an active wetting agent, thoroughly mixed with no more than six gallons of tap water at no less than room temperature (72 F).

3.5.3 Pan Height. The height of the size I drain pan shall be no greater than five inches and that of the size II pan no greater than 5 1/2 inches.

3.5.4 Shock. The drain pan while filled to capacity shall be capable of withstanding without damage the shock of being dropped three times on its bottom surface on a concrete floor from a vertical distance of three feet.

3.6 Components.

3.6.1 Pan Body. When constructed of metallic material, the sheet metal shall be no less than Manufacturer's Standard Gage No. 26 thickness (0.018 inch). When constructed of oil resistant plastic material, the thickness of the material shall assure that the completed pan meets the requirements of this specification. Body joints and seams, when applicable, shall be leakproof and shall comply with the requirements of 3.5 thru 3.5.4. The body shall incorporate a pouring spout designed to provide controlled pouring of the pan's content.

3.6.2 Baffles. The drain pan shall include a baffle to prevent oil from running over the outer edge of the pan during pouring operations.

3.6.3 Handles. The drain pan shall be furnished with two full hand grip handles to assist in lifting and carrying the pan and to permit pushing and pulling the drain pan on level concrete when filled to capacity. One handle shall be located opposite the pouring spout. The other handle shall in no way hinder pouring and shall be capable of being rotated at least 180 degrees from the hinged point to facilitate tilting of the drain pan.

3.7 Corrosion resistance. Metallic components shall be inherently corrosion resistant or shall be plated or treated to resist corrosion in accordance with standard commercial practice.

3.8 Product identification. The drain pans shall be marked for product identification in accordance with MIL-STD-130, and unless otherwise specified (see 6.2), shall include the National Stock Number.

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3.9 Workmanship. Standards shall assure that the drain pans shall have the stability, strength, and characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection(s) are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective materiel, either indicated or actual, nor does it commit the Government to acceptance of defective materiel.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

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(a) Product examination (see 4.5)
(b) Leakage test (see 4.6.1).
(c) Shock test (see 4.6.2)
(d) Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or batch Size	Sample Size
Product Examination (4.5)	1-150	13
Leakage Test (4.6.1)	1-150	13
Shock Test (4.6.2)	1-150	13
Packaging inspection (4.7)	1-150	5

4.5 Examination. Visually and manually examine the pans to determine conformance with the requirements of 3.2, 3.2.1, 3.3, 3.4, 3.5.3, and 3.6.1 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom for defects. Manual examination shall include the operation of movable parts to assure proper functioning. The examination provisions may be applied at the earliest practical point during manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the pan has passed the examinations prescribed by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Leakage test. The drain pan shall be filled to capacity with solution specified in 3.5.2 and then pulled a distance of 15 feet on a concrete floor. The floor shall approximate the surface and condition of a concrete automotive shop floor. After being pulled, the pan shall be permitted to stand, filled with the liquid solution, for not less than 15 minutes. Evidence of any leakage or drop formation at the sides or bottom of the pan, either during or at the end of the 15 minute period, shall be cause for rejection.

4.6.2 Shock test. The drain pans while filled to capacity shall be dropped three times on their bottom surfaces to a concrete floor from a vertical height of three feet. At conclusion of the test, the drain pans shall be examined for evidence of damage and then tested in accordance with 4.6.1. There shall be no evidence of leakage or of damage affecting use.

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4.7. Packaging inspection. Packaging inspection shall be conducted during and after packaging to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C or Commercial, as specified (see 6.2).

5.1.1 Cleaning and drying. Cleaning (process C-1) and drying of drain pans for all levels shall be in accordance with any applicable process and procedure of MIL-P-116 that will not harm the item(s).

5.1.2 Level A.

5.1.2.1 Unit packaging. The unit packaging method shall conform to method III of MIL-P-116. In accordance with 3.2.1 and 3.7, metal pans are corrosion resistant or treated to be corrosion resistant, therefore should need no preservative applied. Each pan shall be protected against damage due to denting chipping or cracking of corrosion resistant finish.

5.1.3 Level C. Level C preservation shall be in accordance with the requirements as specified in MIL-STD-1190.

5.1.4 Commercial. Commercial (COMM) preservation shall be as specified in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, C or Commercial as specified (see 6.2).

5.2.1 Level A. Like quantities of unit preserved drain pans (see 5.1.2.1) shall be packed in an exterior shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The quantity of pans and the gross weight of the container shall be governed by the limitations of the container specification. Closure and strapping of the container shall be in accordance with the applicable container specification. Blocking, bracing and cushioning shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Drain pans preserved as specified (see 5.1.2.1) shall be packed as specified in 5.2.1 except the shipping containers shall conform to PPP-B-640, class 2.

5.2.3 Level C. Drain pans preserved as specified (see 5.1.2.1) shall be packed as specified in accordance with the requirements in MIL-STD-1190.

5.2.4 Commercial. Commerical (COMM) packing shall be as specified in ASTM D 3951.

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5.3 Marking. All interior and exterior markings, whether military or commercial, shall be in accordance with the requirements of MIL-STD-129, in addition to any special or other identification marking required.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The drain pans covered by this specification are intended to be used when draining oil from automotive engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2.1).
(c) First article when required (see 3.1).
(d) Product identification if different (see 3.8).
(e) First article inspection if different (see 4.3)
(f) Quality conformance inspection if different (see 4.4).
(g) Sampling for quality conformance inspection if different (see 4.4.1).
(h) Level of preservation/packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of changes.

6.4 Subject term (key word) listing.

AMSC number
Design
First article
First article inspection
Materials
Performance
Scope

Custodian:

Army - AL
Navy - MC

Preparing activity:

Army - AL

Review activity:

DNA-CS

Project No. 4910-0747

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